

improved by subjecting the model to repeated learning by going through multiple iterations. The product sentiments for new data are predicted using the training model and these are updated in MongoDB.

Data indexing: The River plugin ensures that data changes in MongoDB are immediately available in Elasticsearch, which indexes the data and searches can be performed based on queries built into the system.

Data enrichment: Knowledge bases such as WordNet can help to enrich the information by deriving key characteristics from the metadata. The ontologies of WordNet can help in determining the meaning of the words in the data and mapping them to the metadata feature set.

Data visualisation and insights: Kibana is used to build dashboards for displaying the information in the unstructured data received from the Web, as well as the information received from top contributors to the feature set and the enriched information. It displays information received from top contributors to the feature set and the enriched information from WordNet. Dashboards are also built for inventory and sales information received from internal databases. These dashboards will provide the insights into the existing in-house data, while the external data points to the preferences of the end users and technical experts.

A case study in the retail industry

Intelligent product allocation is a use case for the retail industry, which considers a combination of in-store retail inventory, sales data and social Web analytics to improve decision-making.

The solution for the use case is to have a list of key indicators and a business metric for creating a positive impact. The key measurements are:

- Extracting top selling brands and models along with key features of the product from social media
- Inventory information from internal data

Based on the above measurements, a business metric needs to be arrived for an intelligent and improved product allocation.

An important activity is to determine and finalise the features for extraction. These identified key features are extracted from various blogs and social media. One example is given below.

Types of bag	Pattern type	Closure type	Material
Shoulder bags and totes	Solid	Twist lock	Leather
Satchels and handheld bags	Floral	Zip	Polyurethane
Messenger bags	Dot	Flap	Denim
Crossbody bags	Striped	Magnetic	Nylon
Leather bags	Plaid	Laced	Canvas
	Geometric	Zipper	
		String	
		No zipper	

Table 1: Key features of bags for extraction

Unstructured Data: Few comments on Bags from Social Media and Technical blogs for the key features are given below:

- “Is this item and colour in stock and available? I want to purchase two of them.”
- “I liked this shoulder bag but I want to know if it is pure leather.”
- “I would like to know which bag is on sale. Some pictures display a plain black leather bag, but the other pictures show a bag with a few prints on the sides.”
- “This bag has a lot of room to get all of my stuff into it. I don’t use the shoulder strap because the handles are great and the zippered compartments are big enough.”

Structured Data: From inventory is given in table 2 and table 3

These two tables are examples of the inventory data for the bags available in the store.

Analysis of the multi-structured data

Combining the product preferences, popular characteristics of the bags referred by people in social media and in-store inventory data, the system can come up with a score card to

ITEM	SUPPLIER CODE	COUNTRY_ID	SUPP_PACK_SIZE	STOCK_ON_HAND	UPDATE_DATE-TIME	Colour
100064010	6200	US	8	5	12/5/2015 16:25	Brown
100064110	6200	US	8	1	12/6/2015 16:25	Black

Table 2: In-store inventory information with stocks

ITEM	STATUS	ITEM_DESC	SHORT_DESC	PRICE	CREATE_DATETIME	LAST_UPDATE_DATETIME
100064010	Available	Women's tote bag	Tote bag	150	12/5/2015 15:52	12/6/2015 15:58
100064110	Available	Men's shoulder bag	Shoulder bag	110	12/5/2015 15:52	12/6/2015 15:58

Table 3: In-store inventory information with item description of bags